Simple Numerical Problems

Problem 1: Chip thickness ratio and shear angle

In orthogonal cutting, the uncut chip thickness is 0.20 mm and the chip thickness is 0.50 mm. The rake angle is 10°. Determine the chip thickness ratio and shear angle.

$$r = t_1/t_2 = 0.20/0.50 = 0.40$$

$$\tan(\varphi) = [0.40 \cos(10°)] / [1 - 0.40 \sin(10°)]$$

$$\varphi \approx 22.9°$$

Answer: chip thickness ratio = 0.40; shear angle ≈ 22.9°.

Problem 2: Force components at the rake face

During orthogonal cutting, cutting force F_c = 900 N, thrust force F_t = 450 N and rake angle α = 10°. Determine friction force F, normal force N and coefficient of friction.

$$F = F_c \sin(\alpha) + F_t \cos(\alpha) = 599.4 \text{ N}$$

$$N = F_c \cos(\alpha) - F_t \sin(\alpha) = 808.2 \text{ N}$$

$$\mu = F/N = 0.742$$

$$\beta = \tan^{-1}(\mu) \approx 36.6°$$

Problem 3: Shear force and shear stress

For the cutting conditions of Problem 2, take shear angle φ = 25°, width of cut b = 3 mm and uncut chip thickness t1 = 0.20 mm. Find the shear force and average shear stress.

$$F_s = F_c \cos(\varphi) - F_t \sin(\varphi) = 625.5 \text{ N}$$

$$A_s = b\, t_1 / \sin(\varphi) = 1.420 \text{ mm}^2$$

$$\tau_s = F_s/A_s = 440.6 \text{ N/mm}^2 = 440.6 \text{ MPa}$$

Problem 4: Taylor tool-life equation

A tool has a life of 60 min at 80 m/min and 20 min at 120 m/min. Assuming Taylor's equation $V T^n = C$, determine n and C.

$$V_1 T_1^n = V_2 T_2^n$$

$$n = \log(V_2/V_1) / \log(T_1/T_2) = 0.369$$

$$C = V_1 T_1^n = 362.5$$

Hence the tool-life relation is approximately:

$$V T^{0.369} = 362.5$$

Problem 5: Tool life at a new cutting speed

Using the relation from Problem 4, find tool life at V = 100 m/min.

$$T = (C/V)^{(1/n)} = (362.5/100)^{(1/0.369)}$$

$$T \approx 32.8 \text{ min}$$

Problem 6: Cutting power

If the cutting force is 1200 N and cutting speed is 90 m/min, calculate cutting power in kW.

$$\mathbf{V = 90/60 = 1.5\ m/s}$$

$$\mathbf{P = F_c \times V = 1200 \times 1.5 = 1.80\ kW}$$